

Suggestions to reduce noise pollution

1. Enact legal measures against noise pollution
2. Impose limits on decibel emission
3. Ban sources of noise pollution at public places
4. Create noise-free zones for hospitals, schools, offices and, to a limited extent, residential colonies also
5. The master plan should carefully consider the problems related to noise pollution due to traffic and factories
6. Control noise pollution at the level of technology
7. Make use of silencers compulsory in heavy vehicles and industries generating high decibels
8. Undertake sincere research towards low-noise-producing engines and machines

it is much more difficult with complex sounds such as the spoken word. Specially designed earphones (and headphones) are the most common types of electronic noise-cancellation devices. Devices that do not require earphones are also being developed.

Applications of active noise-cancellation devices

There are a number of applications for active noise-cancellation devices. Noise cancellation requires the sound to be cancelled almost at the source, such as the loudspeaker. That is why the effect works well with headsets, as you can contain the original sound and the canceling sound in an area near your ear. In applications where the sound comes from many areas, such as in a room, it is difficult to cancel the sound from each area. Scientists and engineers are working on solutions.

Headsets. People working near aircrafts or in noisy factories can wear electronic noise-cancellation headsets. Students can wear these headsets to cancel out unwanted noise while listening to their own music during study.

Honda cars. Honda is using noise-cancellation technology in its Japan-only Accord station wagon. A microphone connected to the car stereo system picks up all the sound inside the car, including music or such from the stereo. Then the noise-cancellation system subtracts the sound of the music coming from the stereo and produces noise-canceling sound waves that match the frequency of the unwanted sound. Noise-canceling sound waves are also sent through the stereo speakers, along with the music. This greatly reduces low-frequency vibration noises in the car, without dampening the car's audio system.

Space satellite antennae. Some space satellites have long antennae. If such an antenna starts vibrating wildly, it could throw the satellite out of orbit and out of control. By detecting the waveform or any vibration in the antenna, it can be suppressed in the same way as noise is suppressed. A piezoelectric device creates an electrical signal when it detects a vibration. It also vibrates according to the electrical signal it receives. Putting a piezoelectric device on the antenna can make the antenna vibrate in an opposite phase, thus eliminating the dangerous motion.

Use in apartments. One cool idea to suppress unwanted noise from an adjoining apartment is vibrating your wall out of phase with sound from the next door. For example, if a person in the adjoining apartment has his stereo playing really loud, you may even feel your wall vibrate. By placing devices on the wall, you can detect the wall vibrations and generate vibrations in an opposite phase. This would cancel out the noise coming from the other apartment.

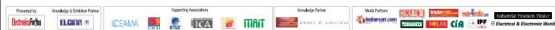
Although there are some devices available that use your wall as a speaker by creating vibrations, these have not been combined yet with detectors and noise-reduction electronics. ●

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